

CONSTRAINED DENSE CORRESPONDENCE GRAPHS FOR ROBUST STRUCTURE-FROM-MOTION TARGETING ENDOSCOPIC VIDEOS

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ABSTRACT

Reconstructing complete 3D geometry from monocular clinical endoscopic videos is challenging due to weak texture, repetitive tissue patterns, and severe illumination artifacts. Although emerging dense matching methods exhibit improved resilience to textureless regions, they often produce abundant spurious correspondences across non-overlapping views, which corrupts the correspondence graph and causes structure-from-motion (SfM) pipelines to fail. In this work, we propose a framework for the construction of dense correspondence graph that leverages explicit temporal locality, parallax-driven geometric constraints, and loop-closure revisiting to enable reliable SfM for monocular endoscopic videos. Instead of exhaustively connecting all frame pairs, the approach effectively suppresses invalid inter-frame correspondences while preserving essential long-range geometric relations critical for stable reconstruction. Combined with illumination-aware masking and SfM initialization adapted to endoscopy, the proposed framework achieves substantial improvements in registration robustness and reconstruction completeness for both phantom and real clinical datasets.

Index Terms—Endoscopy, 3D reconstruction, structure-from-motion (SfM), feature matching, image registration

1. INTRODUCTION

Endoscopy is widely used in minimally invasive procedures to inspect internal organs for diagnosis and intervention [1, 2]. However, monocular endoscopic videos provide only two-dimensional (2D) visual information and inherently lack depth perception, limiting comprehensive understanding of three-dimensional (3D) anatomical structures. Therefore, recovering 3D structure from endoscopic videos has been actively studied [3–7], as it enables intuitive spatial inspection, accurate lesion localization, and improved navigation for computer-assisted and minimally invasive surgeries.

Existing approaches to endoscopic 3D reconstruction can be broadly categorized into SLAM-based, classical

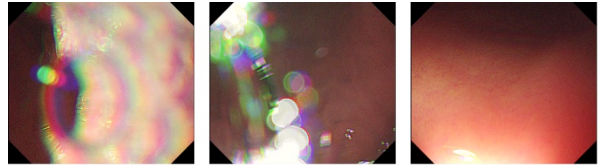


Fig. 1: Images captured from routine endoscopic procedures.

SfM-based, and learning-based methods. SLAM systems emphasize online tracking and real-time performance and have been applied to endoscopic scenarios [10–13]. While suitable for intraoperative guidance, such systems typically prioritize local consistency and may sacrifice global consistency and reconstruction completeness. Learning-based approaches, including supervised methods and self-supervised adaptations of foundation models, have also been explored [14–18]. Supervised methods are constrained by the scarcity of large-scale 3D ground truth in endoscopy, while self-supervised approaches typically rely on joint depth–pose estimation and struggle to recover globally consistent geometry over long sequences. Consequently, achieving reliable high-fidelity reconstruction from long monocular endoscopic videos remains challenging.

In contrast, classical structure-from-motion (SfM) methods, such as COLMAP [9], provide a well-established framework for jointly estimating camera motion and 3D structure from monocular image sequences and remain attractive for high-quality offline reconstruction. However, directly applying off-the-shelf SfM pipelines to clinical endoscopic videos is highly challenging. Endoscopic scenes are characterized by weak and repetitive tissue textures, strong specular highlights, and limited effective parallax due to predominantly forward motion inside narrow anatomical cavities, as illustrated in Fig. 1. These characteristics undermine reliable feature extraction and geometric verification, see Fig. 2a. In particular, as many image pairs share little or no meaningful overlap at all, standard exhaustive pairwise matching, premised on dense graph connectivity, yields geometrically invalid correspondences (Fig. 2b) and results in low registration rates, fragmented point clouds, and frequent pipeline failure.

To alleviate these limitations, prior work has explored image appearance enhancement before reconstruction through for example, the use of indigo carmine (IC) dye [19]

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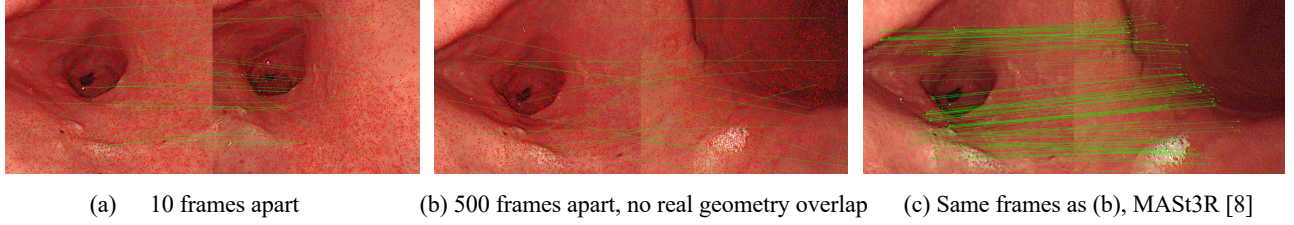


Fig. 2: Feature matching results using (a–b) COLMAP [9] and (c) MAST3R [8] after geometric verification. Green lines denote matched features. COLMAP and MAST3R produce spurious correspondences for distant non-overlapping frames.

or synthesis-based virtual dyeing techniques [20]. While local feature contrast can be enhanced, these approaches introduce additional procedural complexity or rely on learned appearance transformations that may affect clinical applicability. Other methods enhance endoscopic images without dyes [21], but they still rely on conventional SfM pipelines and hence remain vulnerable to the geometric inconsistency arising from indiscriminate graph connectivity across extended sequences

Recent advances in deep dense correspondence estimation [8] have shown remarkable robustness gains for texture-poor scenes. However, exhaustive application of dense matcher over long sequences introduces a proliferation of spurious correspondences between non-overlapping views. These outliers often evade reliable rejection by geometric verification with epipolar geometry, as illustrated in Fig. 2c. Thus, the key challenge lies in how dense correspondences can be organized into a geometrically valid correspondence graph for reliable pose estimation and reconstruction.

Therefore, we formulate endoscopic SfM as a dense correspondence graph design problem under explicit temporal and geometric constraints. We construct a constrained graph that selectively connects frames exhibiting likely valid overlap via temporal locality, parallax-aware keyframe selection, and loop-closure detection while deliberately preserving critical long-range geometric relations. This selective graph design effectively suppresses spurious edges from non-overlapping views and enables robust, globally consistent reconstruction over long endoscopic sequences.

2. PROPOSED METHOD

This section presents our dense correspondence graph construction framework for robust monocular endoscopic structure-from-motion. As illustrated in Fig. 3, the pipeline builds a geometrically valid dense graph whose nodes are video frames and whose edges are dense correspondences constrained by illumination reliability, temporal continuity, and parallax. After image undistortion, a specular and shadow-aware module suppresses illumination-induced artifacts (Section 2.1). Parallax-promoting keyframes are then selected using optical flow to provide stable long-range constraints (Section 2.2). Dense correspondences predicted by MAST3R [8] are integrated through a constrained graph construction strategy (Section 2.3). Finally, camera poses and

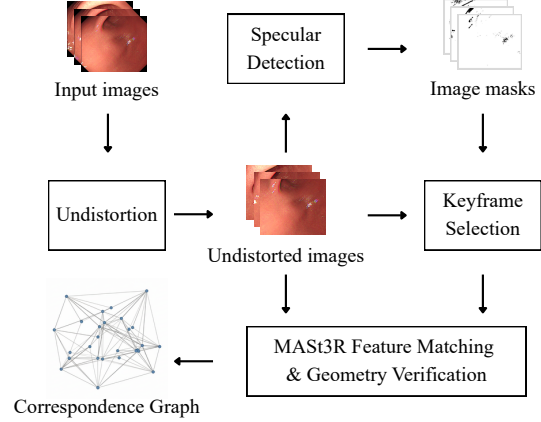


Fig. 3: Overview of the proposed framework.

a 3D point cloud are reconstructed using incremental SfM on the resulting dense correspondence graph (Section 2.4).

2.1. Specular and Shadow-Aware Masking

Endoscopic images are frequently degraded by illumination artifacts, including specular highlights and shadowed regions arising from non-Lambertian tissue surfaces and the co-located camera–light configuration. These regions violate the photometric consistency assumption and introduce unreliable cues for dense motion estimation. To ensure stable geometric reasoning, we define a pixel-wise validity mask by

$$M_{\text{valid}}(x) = M_{\text{tissue}}(x) \cdot M_{\text{specular}}(x) \cdot M_{\text{shadow}}(x), \quad (1)$$

where $x \in \Omega$ denotes a pixel location and Ω is the image domain, M_{specular} and M_{shadow} are non-specular and non-shadow regions, respectively, such that only geometrically valid and photometrically reliable pixels are retained. Specular highlights are detected using [22]. Shadowed regions are conservatively estimated from luminance distribution and local contrast, followed by morphological erosion to preserve essential tissue structures (Fig. 4).

2.2. Optical Flow-Guided Keyframe Selection

Monocular endoscopic videos typically exhibit continuous forward motion and small inter-frame baselines, which lead to weak parallax and unstable scale in correspondence graphs constructed from adjacent frames only. To ensure geometric fidelity with minimal redundancy, we adopt an optical flow-guided keyframe selection strategy that explicitly promote parallax in the dense correspondence graph.



Fig. 4: (a) Input image. (b) Specular mask using [22]. (c) Our mask using Eq. (1).

Given the current frame I_t and the most recent keyframe I_k , bidirectional dense optical flow is estimated using RAFT [23]. Flow consistency and motion statistics are evaluated within the valid tissue mask M_{valid} . We define a tissue overlap ratio $\rho(t, k)$ as the fraction of valid pixels satisfying forward-backward flow consistency. τ_p , τ_r , τ_{ssim} , and Δ_{min} are fixed thresholds. The optical flow-guided keyframe selection mechanism is then triggered by four conditions. First, I_t is selected when $\rho(t, k) > \tau_p$, indicating insufficient geometric overlap with I_k . Second, I_t is retained when $div_r(t, k) > \tau_r$, that is, the scale change is observed through radial optical flow to enhance triangulation quality. In addition, candidates are further filtered by structural similarity $SSIM(I_t, I_k) < \tau_{ssim}$ and temporal interval $|t - k| \geq \Delta_{min}$ to avoid redundancy.

2.3. Dense Correspondence Graph Construction

Conventional exhaustive dense matching introduces spurious correspondences across non-overlapping views, corrupting the graph and destabilizing SfM. We therefore construct a constrained graph using temporal, parallax, and loop-closure priors to suppress invalid edges while preserving long-range geometric relations. The graph topology comprises three edge types defined as follows:

$$E_{local} = \{(I_i, I_j) \mid 1 \leq j - i \leq \delta\}, \quad (2)$$

$$E_{keyframe} = \{(K_i, K_j) \mid K_i, K_j \in \mathcal{K}, 1 \leq j - i \leq \delta_k\}, \quad (3)$$

$$E_{loop} = \{(I_i, I_j) \mid s(I_i, I_j) \geq \tau_{loop}, |i - j| > \Delta_{loop}\}, \quad (4)$$

where δ and δ_k are sliding window sizes, all frames I_i form the set V , $\mathcal{K} \subset V$ is the set of selected keyframes K_i , $s(\cdot)$ denotes CosPlace [24] descriptor similarity, and τ_{loop} , Δ_{loop} are fixed thresholds.

For local edges E_{local} , each frame I_t connects to subsequent frames within a sliding window $\{I_{t+1}, \dots, I_{t+\delta}\}$, forming short-range correspondences that ensure dense yet reliable correspondences while avoiding mismatches across unrelated tissue regions. In addition, keyframe edges $E_{keyframe}$ are matches established within a sliding window $\{K_{i+1}, \dots, K_{i+\delta_k}\}$ for each keyframe K_i that stabilizes reconstruction by mitigating scale drift from short-baseline connections. Lastly, loop-closure edges E_{loop} retrieve local matching pairs of the same regions from distant frames using CosPlace [24]. Frame pairs with high descriptor similarity and sufficient temporal separation are connected to form edges that are not captured by local or keyframe edges.

The final correspondence graph E_{final} is the union of these three edges:

$$E_{final} = E_{local} \cup E_{keyframe} \cup E_{loop}, \quad (5)$$

where dense features are further matched within each edge of E_{final} using MAST3R [8]. Since MAST3R predicts pixel-wise correspondences, directly using all points is computationally prohibitive. We uniformly subsample correspondences on a regular grid and discard points with confidence score below a threshold C_{TH} . The sampled correspondences are then verified using RANSAC-based essential matrix estimation.

2.4. Incremental SfM with Adapted Initialization

All geometrically verified edges in the dense correspondence graph are stored in a COLMAP-compatible database. Camera poses and point cloud are then reconstructed using the COLMAP mapper and bundle adjustment. Due to predominantly forward camera motion, initial view pairs often exhibit small triangulation angles. To accommodate this characteristic, we relax the minimum triangulation angle threshold θ_{min} during initialization, allowing reconstruction from small-baseline pairs. Combined with the constrained pair selection in our graph, this adapted initialization enables coherent reconstruction in challenging endoscopic scenes.

3. EXPERIMENTS AND DISCUSSION

3.1. Data Acquisition

Clinical endoscopic videos were acquired using an OLYMPUS GIF-H290 monocular endoscope during routine gastroscopy procedures. RGB videos were recorded at 30 fps using a Sony HVO-550MD medical video recorder. To reduce gastrointestinal peristalsis, standard antispasmodic protocols were applied. After removing regions containing protected health information, each frame has an effective resolution of 1005×864 pixels. This study was approved by the Institutional Review Board (IRB) of National Taiwan University Hospital, Approval No. 202407132RINC.

For quantitative evaluation, we additionally employ the C3VDv2 phantom dataset [25, 26], which provides ground truth geometry for quantitative evaluation. The dataset contains multiple colon geometries (c0–c2), anatomical segments with varying surface textures (t1–t4), and imaging conditions (v1–v3) with endoscopic-specific artifacts such as mucus, debris, and water droplets, enabling realistic assessment under challenging visual conditions.

3.2. Implementation Details

All experiments were conducted on a server with an NVIDIA RTX 4090 GPU (24 GB VRAM). The window sizes δ and δ_k were both set to 5, $\tau_p = 0.6$, $\tau_{ssim} = 0.8$, and $\tau_{loop} = 0.85$. Predicted correspondences were subsampled on an 8-pixel grid stride with $C_{TH} = 0.5$. During incremental SfM, the minimum triangulation angle was relaxed to $\theta_{min} = 2.0^\circ$.

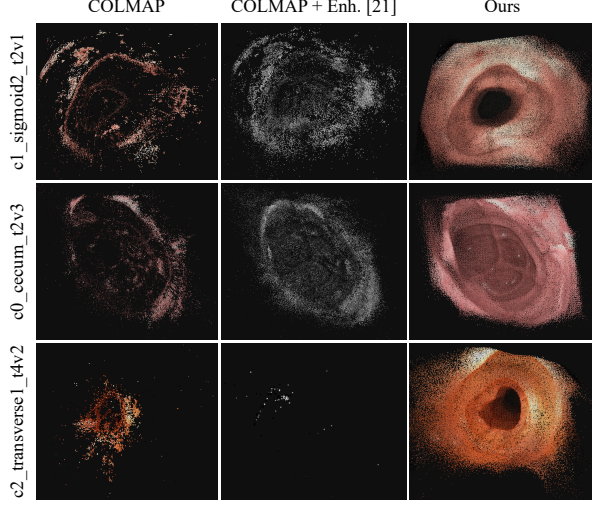


Fig. 5: Reconstructed point clouds for three C3VDv2 sequences [25, 26].

3.3. Results of 3D Reconstruction

3.3.1. Phantom Dataset

Quantitative results on five representative sequences of C3VDv2 are summarized in Table 1, following the evaluation protocol of [27]. Reconstruction errors are reported as point-to-triangle distances on surveyed regions (directly observed surfaces) and extended regions (including neighboring points within a 20 mm radius), using MedAE (median absolute error), MAE (mean absolute error), and RMSE.

The proposed method consistently outperforms vanilla COLMAP [9] and previous method [21] across all metrics. For instance, in challenging sequences such as *c2_ascending_t4v1*, COLMAP registers only 73/292 frames, resulting in highly incomplete reconstructions. In contrast, our approach achieves full registration and significantly lower errors, demonstrating superior robustness.

Qualitative comparisons are shown in Figs. 5 and 6. As observed in Fig. 5, our pipeline yields denser and more coherent point clouds with fewer structural gaps. Previous methods fail to reconstruct sequences affected by water droplets on the lens (e.g., *c2_transverse1_t4v2*), whereas our approach successfully preserves anatomically meaningful structures. Trajectory comparisons in Fig. 6 further show that our method produces smoother and more stable camera trajectories, while baselines exhibit fragmentation or drift due to unreliable correspondence estimation.

3.3.2. Real Dataset

For real clinical endoscopic videos where ground truth geometry is unavailable, we assess the performance by visual coherence, anatomical plausibility, and registration robustness. As shown in Fig. 7, our method produces anatomically plausible and consistent reconstructions despite non-rigid tissue motion, weak texture, and illumination variation. In contrast, the baseline methods frequently fail for long-term tracking and produce fragmented reconstructions.

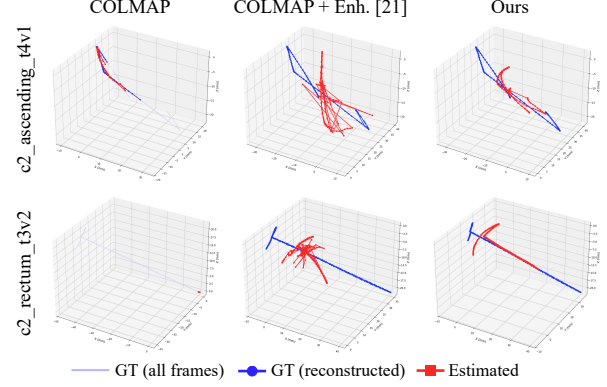


Fig. 6: Trajectory estimation results for two sequences of different colon parts obtained from C3VDv2 [26].

To show quantitative robustness, Table 2 reports frame registration rates on seven clinical sequences. The proposed method achieves near-complete registration on six sequences and superior performance over the baseline methods. For the most degraded case, Dataset 5 contains a prolonged segment where the endoscope travels in close contact with the gastric wall, producing sustained textureless regions, as shown in Fig. 1. This significantly reduces the number of reliable correspondences, causing partial track breaks and limiting the overall registration rate. Despite this, our method achieves anatomically meaningful geometry (Fig. 7) and 51.9% registration rate, substantially outperforming COLMAP (5.4%) and COLMAP + Enh. (8.9%).

3.4. Ablation Study

To isolate the impact of correspondence graph design, we compare three graph topologies using the same MAST3R matcher: (i) a temporal-local graph using sequential matching with window size 10, (ii) an unconstrained graph with exhaustive matching, and (iii) the proposed constrained graph.

As shown in Fig. 8, the temporal-local graph suffers from drift and incomplete reconstruction due to insufficient long-range constraints. The unconstrained dense graph introduces excessive spurious connections that compromise pose estimation and reconstruction fidelity, while our proposed strategy balances temporal continuity and geometric validation, yielding more accurate point clouds and stable trajectories. In addition, this reduces the number of MAST3R inference calls from $O(N^2)$ to $O(N \cdot \delta + |\mathcal{K}| \cdot \delta_k + |E_{loop}|)$, enabling our method to handle 1,201 frames for Datasets 3 and 5. In contrast, the exhaustive dense matching method becomes computationally intractable beyond 400 frames.

To show the individual contribution of each component of our design, we perform a component-wise ablation on Dataset 1. The results in Table 3 further confirm that keyframe edges achieve the largest gain by resolving scale drift of 3D reconstruction, while loop-closure edges retrieve local matching pairs of the same regions from distant frames to perfect the reconstruction completeness.

Table 1: Point-to-triangle reconstruction errors across different anatomical segments for the C3VDv2 dataset. Results are reported as **MedAE / MAE / RMSE (mm)**. N denotes the total number of frame for each sequence.

Method		Sequences (Number of Input Images)				
		c0_cecum_t2v3 (<i>N</i> = 595)	c1_sigmoid2_t2v1 (<i>N</i> = 400)	c1_transverse2_t1v3 (<i>N</i> = 517)	c2_ascending_t4v1 (<i>N</i> = 292)	c2_transverse1_t4v2 (<i>N</i> = 382)
Ext.	COLMAP	7.93 / 9.28 / 10.79	5.57 / 7.90 / 13.14	4.24 / 6.12 / 11.67	10.88 / 11.32 / 12.02	6.01 / 8.76 / 29.51
	COLMAP + Enh. [21]	8.02 / 8.85 / 10.74	5.46 / 7.64 / 15.08	3.90 / 6.36 / 11.34	8.66 / 10.17 / 15.36	14.76 / 15.08 / 18.40
	Ours	7.90 / 8.41 / 9.68	5.21 / 6.44 / 8.21	4.01 / 6.00 / 8.54	8.65 / 8.79 / 10.05	6.77 / 8.18 / 10.80
Sur.	COLMAP	7.93 / 9.28 / 10.79	5.57 / 7.91 / 13.34	4.30 / 6.17 / 11.69	10.88 / 11.32 / 12.02	6.74 / 9.28 / 29.73
	COLMAP + Enh. [21]	8.02 / 8.86 / 10.74	5.47 / 7.69 / 15.52	3.97 / 6.40 / 11.36	8.69 / 10.22 / 15.53	14.76 / 15.08 / 18.40
	Ours	7.90 / 8.41 / 9.68	5.21 / 6.44 / 8.22	4.04 / 6.02 / 8.55	8.65 / 8.79 / 10.05	6.81 / 8.23 / 10.87

Table 2: Quantitative evaluation of image registration robustness on real datasets. Results are reported as the number of registered images and the corresponding registration rate (%).

Method	Dataset 1 ($N=601$)	Dataset 2 ($N=801$)	Dataset 3 ($N=1201$)	Dataset 4 ($N=651$)	Dataset 5 ($N=1201$)	Dataset 6 ($N=651$)	Dataset 7 ($N=501$)
COLMAP	298 (49.6%)	385 (48.1%)	645 (53.7%)	378 (58.1%)	65 (5.4%)	42 (6.5%)	35 (7.0%)
COLMAP + Enh. [21]	323 (53.7%)	63 (7.9%)	768 (63.9%)	503 (77.3%)	107 (8.9%)	156 (24.0%)	38 (7.6%)
Ours	599 (99.7%)	799 (99.8%)	1199 (99.8%)	651 (100.0%)	623 (51.9%)	645 (99.1%)	500 (99.8%)

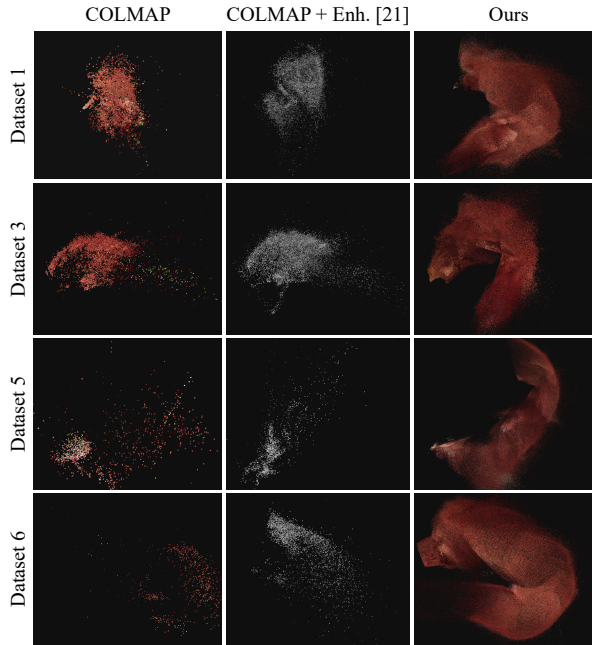


Fig. 7: Qualitative reconstructions for real clinical endoscopic sequences.

4. CONCLUSION

In this paper, we have described a dense correspondence graph framework for robust structure-from-motion that integrates illumination-aware masking, flow-guided keyframe selection, and constrained graph construction for monocular endoscopic videos. By leveraging the temporal and geometric constraints to selectively control graph topology, the proposed method effectively suppresses

Table 3: Component-wise ablation on Dataset 1 ($N=601$).

E_{local}	$E_{keyframe}$	E_{loop}	Registration Rate
✓	✗	✗	2.2%
✓	✓	✗	98.5%
✓	✓	✓	99.7%

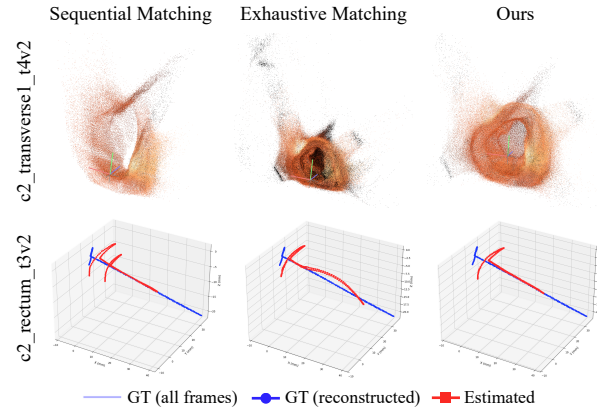


Fig. 8: Ablation on trajectory and point cloud for C3VDv2 dataset using different graph topologies.

spurious correspondences from non-overlapping views and stabilizes long-range reconstruction in challenging endoscopic scenes. Experiments on both phantom and real clinical datasets demonstrate substantial improvements in frame registration, reconstruction completeness, and trajectory stability over existing SfM pipelines. These advancements highlight constrained dense correspondence graph design as a key enabler for robust endoscopic 3D reconstruction and supports downstream clinical applications requiring reliable anatomical geometry.

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